

Medhansh Garg

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EDUCATION

University of Illinois, Urbana-Champaign

Aug. 2023 – May 2027 (Expected)

- B.S. in Computer Engineering | Relevant Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Databases

SKILLS

- **Languages:** TypeScript, Python, Java, C, C++, SQL
- **Frameworks & Web:** React, Next.js, Node.js, FastAPI, Flask, Spring Boot, Prisma, SQLAlchemy, REST APIs
- **Data & Cloud:** PostgreSQL, SQLite, Supabase, Docker, Kubernetes, AWS, Azure, GitHub Actions CI/CD, Vercel, nginx, Git, Linux
- **ML & CV:** PyTorch, MediaPipe, OpenCV, YOLOv8, NumPy, pandas

EXPERIENCE

Hopcharge | *Software Engineering Intern*

Jun. 2026 – Aug. 2026

- **Ad Engine** — Architected a closed-loop marketing pipeline (**Next.js 16**, **React 19**, **Prisma/Postgres**) that generates, reviews, publishes (**Meta/YouTube**), and learns from ad creatives — scaling output from 2–3/week to **3/day** ($\approx 8\times$)
- Made every vendor (LLM, image/video, publishers) hot-swappable via one env var; scheduled jobs on a **pg-boss** (Postgres) queue with a serverless **Vercel Cron** fallback; a per-platform feedback loop (Meta CPL, YouTube engagement) drives ideation
- **VOLT.CV** — Built a **FastAPI** HR dashboard automating hiring end-to-end: resume parsing (3-tier OCR: **pdfplumber** → **PyMuPDF** → **Tesseract** + **spaCy**), Gmail reply classification, and one-click bulk interview scheduling
- Secured it with database-level **Row-Level Security** tied to Google identities over **Supabase/PostgreSQL**; shipped a cross-platform desktop build via **PyInstaller** + **GitHub Actions** CI targeting 4 OS/architectures
- **Grid Ops** — Replaced 5 hand-run scripts with a **FastAPI** + **React (Vite/TS)** service mapping demand, churn, forecasting, and charging-pod allocation across **3 Delhi-NCR metros** (~ 60 hexes), deployed behind **nginx** on an ARM VM

CS 124 Honors | *Lead Web Developer*

Aug. 2025 – Present

- Built and lead the CS 124 platform (**Next.js**, **Supabase**) serving **300+ students** and **30+ staff** each semester, with **CILogon** SSO (replacing an insecure prototype) and a 6-role least-privilege system granting temporary role-view access without permanent escalation
- Routed all public pages through a server-side API layer so no DB keys reach the client; gated dev→main on a **Vitest** + **Playwright** E2E suite (isolated test tables, real JWT sessions) in **GitHub Actions** CI
- Automated student/staff, grading, and event management (batch grading, CSV import, sprint tracking), cutting manual overhead/error; lead an **11-person** dev team via weekly meetings and PR reviews

Care Health Insurance | *Cybersecurity Intern*

Jun. 2025 – Aug. 2025

- Ran external recon uncovering **60+ unauthenticated internet-exposed services**; reverse-engineered **WAF request-signing** into a **Burp Suite** replay PoC and bypassed **SSL pinning** + root-detection via **Frida** + **JADX**

IoT++ | *Software Engineering Intern*

Jun. 2024 – Aug. 2024

- Built a real-time fire/smoke detection pipeline (**YOLOv8**): **+62%** accuracy, **13 $\times$** throughput via **pruning** + **tiling**; deployed **NPU**-optimized inference on Orange Pi, containerized with **Docker** + **Kubernetes** on **Azure**

PROJECTS

ASL Adaptive Tutor

github.com/HackOverflow404/SignLanguageLearning

- Real-time computer-vision tutor for American Sign Language: trained a multi-task **triplet-embedding** grader (**PyTorch**) over **MediaPipe** pose landmarks, reaching **81.2% top-1** / **96.5% top-5** across 60 signs — **2.5 $\times$** a DTW baseline
- Benchmarked 4 pose backends across 1,874 clips; built a live-webcam grader with per-parameter (handshape/movement/location) feedback and an **SM2 spaced-repetition** scheduler with contrastive minimal-pair drills

AlberFlowy

github.com/HackOverflow404/AlberFlowy

- Reverse-engineered WorkFlowy's private API for full tree CRUD via a **Node.js CLI**; built a **C++** Albert plugin with caching, async sync, and **Google OAuth2** automation
- Benchmarked on a 150+ item production vault: live plugin queries return in **p50 0.7ms** / **p95 1.7ms** at a **100%** cache hit rate; caching cuts repeat syncs from ~ 265 ms to **<1ms**, and the sync engine parses 10k-node trees in ~ 3 ms

RemoteCam

github.com/HackOverflow404/RemoteWebcam

- Built a cross-platform **WebRTC** webcam streaming system with **Firestore** signaling; **QtPython** Linux client + **Next.js** mobile **PWA** for real-time camera sharing

LEADERSHIP & INVOLVEMENT

Illinois MicroTech | *External Vice President* | *EOH Project Lead*

Oct. 2025 – Present

- *External VP & EOH 2026 Project Lead*: led the Engineering Open House showcase for **20,000+ attendees**